Enrollment No.: _____

Darshan UNIVERSITY

योगः कर्मसु कौशलम्

Darshan Institute of Computer Application

MCA | Sem-3 | Winter-2024

Course Code	:	2305CS302	**Date**	:	16-10-2024
Course Name	:	Introduction to Advanced Technologies	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	Define Cloud Computing. List various cloud service providers available in the Industry.	**4**
	(B)	Write the characteristics of cloud computing.	**3**
		OR	
		Define virtualization. List out types of Virtualizations. Explain anyone in detail.	
	(C)	Explain the types of clouds in detail.	**7**
		OR	
		Explain the cloud service model (IAAS, PAAS, SAAS).	
Q.2	**(A)**	Compare Augmented Reality and Virtual Reality with minimum two applications.	**4**
	(B)	Write Visualization Techniques for Virtual Reality (VR).	**3**
		OR	
		Define Mixed Reality with an example.	
	(C)	Determine challenges with AR and VR.	**7**
		OR	
		Determine at least seven practical applications of AR and VR in brief.	
Q.3	**(A)**	What is machine learning? How it works?	**4**
	(B)	Describe Training, Testing and Feature with respect to ML.	**3**
		OR	
		Describe Task, Algorithm and Label with respect to ML.	
	(C)	State supervised, unsupervised and reinforcement learning with suitable example.	**7**
		OR	
		Draw and write about machine learning life cycle with diagram.	

Q.4	**(A)**	Explain data science pipeline.	**4**
	(B)	Explain importance of data science.	**3**
		OR	
		Explain core competencies of data science.	
	(C)	Explain python ecosystem for data science.	**7**
		OR	
		Explain groupby() function in pandas with example.	
Q.5	**(A)**	Briefly state the concept of decentralization.	**4**
	(B)	Write altcoins in brief.	**3**
		OR	
		Describe fundamental working of Proof of Stake consensus algorithm.	
	(C)	Determine types of blockchain.	**7**
		OR	
		What is cryptocurrency? Determine types of forking.	

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